

Extended Abstract

Motivation The objective is to show the effectiveness of integrating optical tactile feedback into the state space of a visuomotor diffusion policy model to safely manipulate soft fruit with robotic arms. Vision has been shown to be highly effective for robotics applications as it provides rich state information for the robot. However, when it comes to manipulating soft and fragile fruits such as strawberries or blackberries, vision alone can be insufficient and lead to potential damage during manipulation. This is an important contribution as automated harvesting in the agriculture sector is a key technology that can help address the growing global population and increasing shortages in the agricultural labor force.

Method The method is relatively simple and straightforward. Multiple diffusion policies will be trained and tested on a real robot arm where a pick and place task with a strawberry is to be completed. The policies will use various observations, specifically comparing the difference in using only external vision versus using external vision and optical tactile sensor feedback.

Implementation For implementation, I used an xArm7 7 degree of freedom robot arm, along with two types of grippers. The first gripper is the UFACTORY parallel jaw gripper and is the baseline that uses only external vision in the policy. The second gripper is a modified parallel jaw gripper that utilizes two DenseTact Optical Tactile Sensors that will provide two additional RGB images in the policy. The policies both utilize 100 human tele-operated demonstrations of a pick and place for real strawberries for 2D diffusion policy training. I then trained a baseline policy with just external vision using the parallel jaw gripper and then trained a policy using just external vision with the DT gripper, and then finally a policy with the DT gripper using external vision and the optical tactile feedback.

Results I found that both grippers that trained on just external vision had a success rate of 60 percent, while the DT gripper policy with external vision and the optical tactile feedback only showed a 5 percent success rate during testing.

Discussion The results make sense because in literature they used 200+ demonstrations for real world policies, whereas I only used 100 demonstrations (due to time constraints). This is likely the reason for the baseline policy only achieving 60 percent success rate. In terms of the low success rate for the DT gripper with sensor feedback, I believe it did not work as well because the images don't provide enough crucial information in the image encoding space, and it only serves to distract the policy. Since this feedback was only raw RGB images, the policy struggled to find the correlation between unique RGB pattern of the DenseTact sensor and the other RGB images.

Conclusion In summary, I was able to get somewhat working baseline policies but discovered that they did not perform as well as they could have been due to insufficient human demonstrations. I also discovered that adding raw DenseTact Optical Tactile Sensors to the policy input did more harm than good, leading to worse policy performance of the pick and place task.

Integrating Optical Tactile Feedback into Diffusion Policy for Soft Fruit Manipulation

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Abstract

Manipulating soft and fragile fruits such as strawberries with robotic arms remains challenging: while vision provide rich state information, it alone can lead to unintended post-handling damage. In my work, I investigate the integration of optical tactile feedback into a 2D visuomotor diffusion policy for a pick-and-place strawberry task. Using an xArm7 robot, I compare three policies trained on 100 human tele-operated demonstrations: (1) a baseline policy with a standard UFACTORY parallel-jaw gripper and vision only; (2) a policy using the same gripper but still relying solely on vision; and (3) a policy combining external vision with raw RGB images from two DenseTact optical tactile sensors mounted on a modified parallel-jaw gripper. In testing, both vision-only policies achieved a 60 percent success rate, whereas the policy incorporating raw tactile images achieved only a 5 percent success rate under identical conditions. These results suggest that, given limited demonstrations, naive inclusion of high-dimensional tactile imager can distract the diffusion model rather than improve it. I conclude that effective tactile integration in diffusion-based manipulation requires more informative sensor representations and/or larger demonstration datasets.

1 Introduction

In 2005, an estimated 38% of the world’s land area was occupied by agricultural needs alone Schneider et al. (2011). With the global population projected to reach 8.5 billion by 2030 United Nations (2015), it is estimated that without significant technological development in the agricultural sector, 50% of all land will need to be devoted to agriculture to maintain food consumption levels. Furthermore, by 2050, an estimated 70% increase in food production will be needed to support a projected 9.1 billion world population Food and Agriculture Organization of the United Nations (2009). To meet these growing needs, substantial improvements in agricultural technology have been made, specifically the development and application of machinery and automation. These advances have allowed for lower production costs, reduced time spent on strenuous labor, increased fresh produce quality, and more Edan et al. (2009).

Among the various activities involved in the agricultural cycle, crop harvesting is a time-consuming and labor-intensive task that constitutes a significant portion of farm expenditure. For example, in the blackberry industry, harvesting hours account for 40-50% of the total man-hours in maintaining and harvesting a crop Hall and Funt (2017). As a result, in one study, the cost of harvesting contributed to 56% of the total cost of bringing a blackberry to market Bolda et al. (2018). These considerations have pushed automated harvesting to the forefront of extensive research within agricultural robotics Navas et al. (2021).

Numerous methods for automated harvesting have been developed for various crop types, such as apples Bu et al. (2020); Lv et al. (2019); Silwal et al. (2016), tomatoes Kondo et al. (2008);

Kounalakis et al. (2021); Ling et al. (2004), strawberries Xiong et al. (2019); Yamamoto et al. (2014); Yu et al. (2019), onions Omar et al. (2018), and more, which have been successfully implemented. For example, Brown et al. (1996) developed a mechanized blueberry harvester capable of harvesting 68% blueberries without post-harvest damage, compared to an average of 77% for traditional hand harvesting.

Over the past decade, deep visuomotor learning has enabled robots to gain manipulation skills directly from using raw camera observations and proprioception, which has replaced hand-engineered perception-control pipelines with end-to-end neural policies Levine et al. (2016). While such approaches can imitate expert demonstrations, their autoregressive action decoders can struggle with multimodal behaviours and high-dimensional continuous control. Diffusion-based policies address these challenges by treating action generation as a conditional denoising process: during inference, the network iteratively refines a noisy action trajectory towards one that is consistent with the current visual context, naturally capturing multimodality and offering good training stability. The development of diffusion policy motivates the exploration for adding tactile sensor feedback to see if it can improve robot policies for soft fruit handling.

2 Related Work

My project will be focusing on the area of imitation learning to teach robot policies. Specifically, I will be utilizing diffusion policy as my imitation learning method, as presented by Chi et. al (2023). Diffusion-based visuomotor policies utilize Denoising Diffusion Probabilistic Models (DDPM) as a way to learn high dimensional policies Ho et al. (2020). In Chi et. al's work, they utilize 2D visual information and robot proprioception for the observation state space. They conducted various experiments in both simulation and real-world to evaluate their diffusion policy architecture performance, including a push T task, 6-Dof mug flipping, sauce-pouring and spreading, etc. These experiments showcased the power of diffusion policy through both the speed and success rate of the trained policies.

With regards to optical tactile sensing for manipulation for agriculture, Han et. al utilize optical tactile sensing to manipulate fruits Han et al. (2024). However, their approach focuses on harder and stiffer fruits that are less susceptible to post-manipulation damage such as apples, lemons, and plums. The authors have yet to demonstrate their method can be applied to extremely fragile fruits like blackberries and strawberries.

In Mahmoudi et al. (2024), the authors do a survey of the applications of imitation learning in the agricultural space and there has been no work so far that has been done on applying diffusion policies in this space. This is a driving factor in pursuing the above proposed research direction, as it is both novel and important to the field of robot learning and agricultural robotics as a whole.

3 Method and Experimental Setup

I followed a three-step methodology: First, demonstration collection: using a 3Dconnexion Space-Mouse, an operator tele-operated the robot through 100 pick-and-place demonstrations with a standard UFACTORY parallel-jaw gripper fitted with silicone pads and a further 100 demonstrations with the same gripper after both fingers were replaced by DenseTact optical tactile modules. This yielded two separate 100-trajectory datasets (200 demonstrations total). Second, policy training: for each dataset we trained an image-conditioned 2-D diffusion policy on an RTX 4090 GPU. The network observed the previous eight time-steps, produced a four-step action sequence, and was optimized to predict eight future steps in joint-position space. Training used a learning rate of $1e-4$, weight decay of $1e-6$, and 100 discriminator iterations per generator update. Third, evaluation: each trained policy was deployed in 20 autonomous pick-and-place trials while running inference at 10 Hz within ROS 2 Humble. Sensor streams (external vision and, for the DenseTact gripper variant, high-resolution deformation images) were fused using approximate-time synchronization, and the strawberry's initial pose was randomly varied across trials. Success or failure was recorded for every rollout to quantify task performance. Figures 1 and 2 show the set up, Figure 3 and 4 show the grippers used.

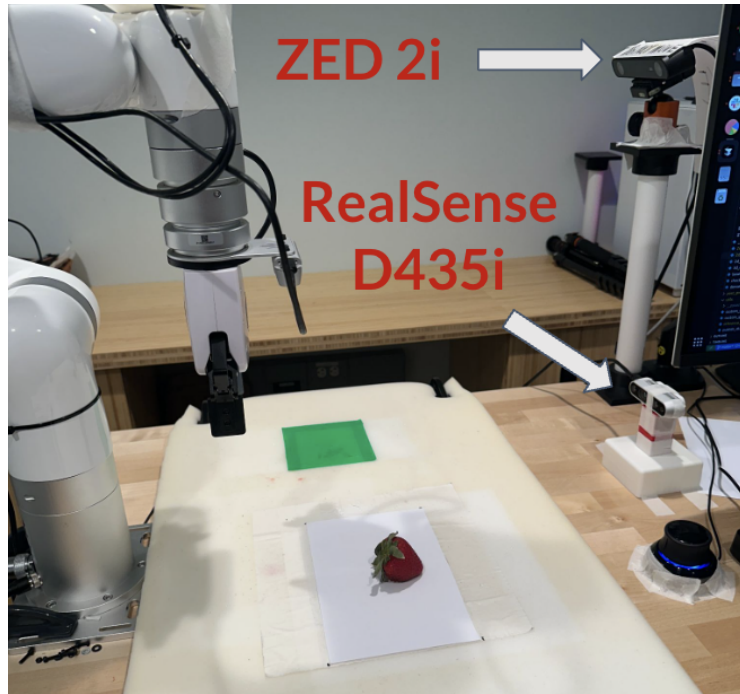


Figure 1: Baseline Gripper Setup.

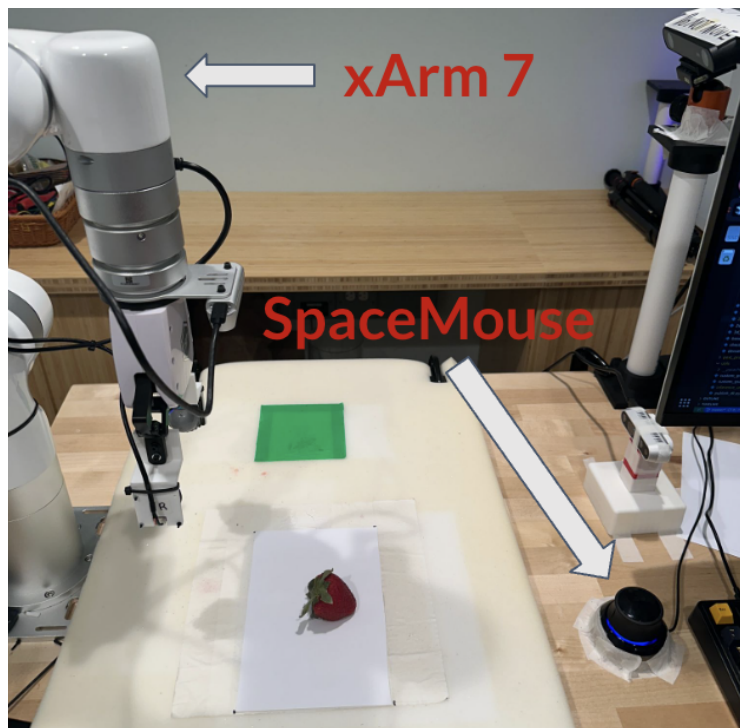


Figure 2: DenseTact Gripper Setup.

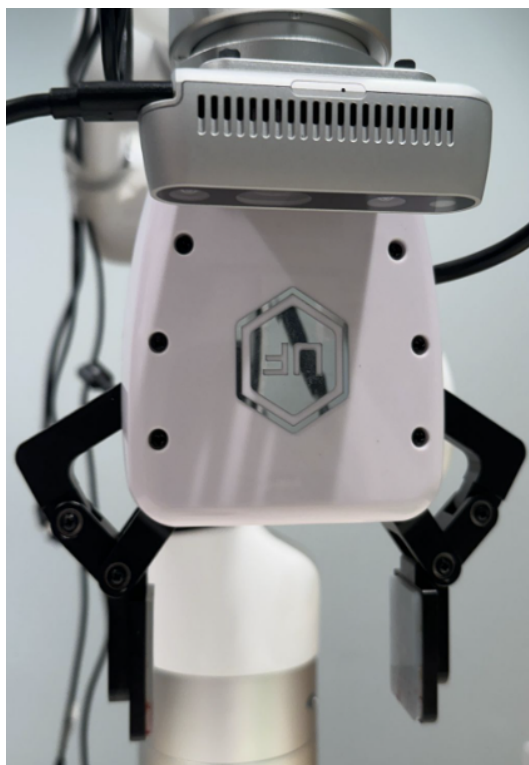


Figure 3: Baseline Gripper.

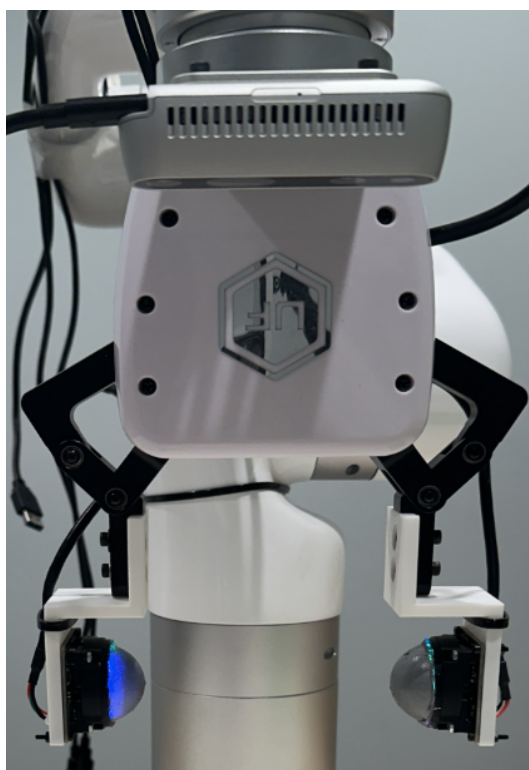


Figure 4: DenseTact Gripper.

4 Results

4.1 Quantitative Evaluation

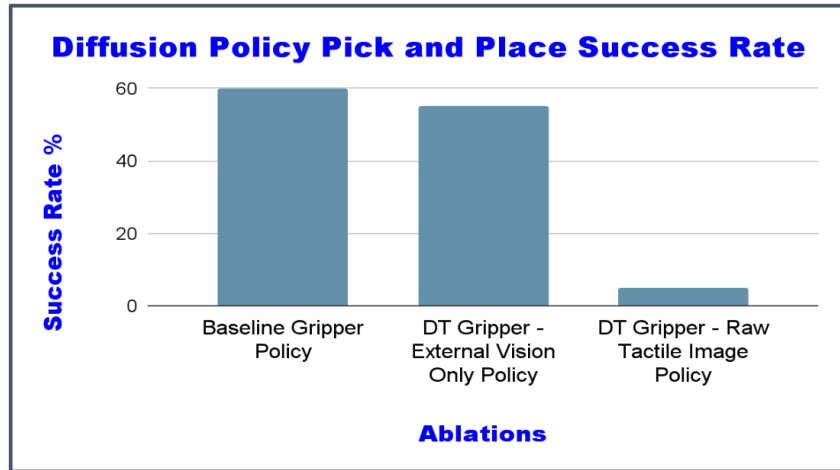


Figure 5: Visualization of results.

The above figure shows the qualitative results of the experiments.

4.2 Qualitative Analysis

The below figure depicts a strawberry post pick and place using the baseline gripper:



Figure 6: Strawberry Damage from Baseline Gripper

The baseline gripper had many instances of visible damage done to the strawberry as shown in the above image. The DT gripper did not show any damage done to the strawberries.

Below shows a comparison between no contact and contact for a raw DenseTact optical sensor image.

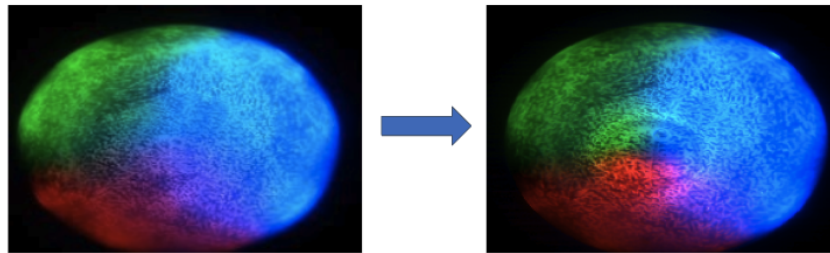


Figure 7: DenseTact Sensor Raw Image Comparison

5 Discussion

The results make sense because in literature they used 200+ demonstrations for real world policies, whereas I only used 100 demonstrations (due to time constraints). This is likely the reason for the baseline policy only achieving 60 percent success rate. In terms of the low success rate for the DT gripper with sensor feedback, I believe it did not work as well because the images don't provide enough crucial information in the image encoding space, and it only serves to distract the policy. Since this feedback was only raw RGB images, the policy struggled to find the correlation between unique RGB pattern of the DenseTact sensor and the other RGB images.

With regards to the damage, the results I saw made sense because the baseline gripper mechanically isn't as compliant as the DT gripper, thus damage is inevitable over time as pick and place is performed over and over. Additionally, the position of the strawberry with respect to the gripper will affect the damage the jaw does to the strawberry.

6 Conclusion

In summary, I was able to get somewhat working baseline policies but discovered that they did not perform as well as they could have been due to insufficient human demonstrations. I also discovered that adding raw DenseTact Optical Tactile Sensors to the policy input did more harm than good, leading to worse policy performance of the pick and place task.

For future work, I hope to collect more demonstrations and improve the baseline performance. Then I hope to utilize calibrated sensors that can provide force feedback, which can be included in the model observation channel, this will hopefully provide more directly useful information for the diffusion policy to learn.

7 Team Contributions

- **Group Member 1:** Everything.

Changes from Proposal N/A

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A Additional Experiments

N/A

B Implementation Details

N/A